

Lesson 4.3 Build a Chatbot to Test our Message

Location: Classroom

Grade Level: High school

Time: 50 minutes

Driving Question

- How can we test our science communication message?

Lesson Synopsis

In this lesson, students will test the science communication plan they have created in Module 4.2. They will do so by building a chatbot using generative AI (GenAI) to represent their target audience. They will then test the communication plan with the chatbot. Students will gather feedback from the chatbot to help them refine their products in Module 4.4.

What students will figure out:

- Effective communication requires feedback and iteration—or reciprocity in communicating with our audience. A message that seems strong to the communicator may not land the same way with the audience, and testing is essential.
- Different audiences interpret the same message differently, depending on their background, interests, and values. By building chatbots to role-play these audiences, students see how communication must be adapted to meet those needs.

LEARNING OUTCOMES

By the end of this investigation, students will be able to...	You can assess this using...
1. Create a text-based chatbot using generative AI to test their messages.	Chatbot prompts in the Investigate section.
2. Collect feedback from the chatbots about their science communication plans.	Science journal in the Navigate section.

Building towards NGSS and AI Literacy:

- **AI Literacy's Big Idea #4: Natural Interaction.** Describe several approaches to natural language processing (e.g., how a chatbot can work using keywords, templates, or machine learning like GenAI). Students will use prompts to develop their generative AI chatbots to test their science communication messages.

LESSON OVERVIEW

Section	Time	What Happens
Navigate	10 minutes	Students get an introduction to the lesson, which will focus on creating a chatbot to simulate their communication audience.
Investigate	20 minutes	Students create prompts (with ChatGPT/Gemini) to create the targeted audience chatbot.
Investigate	15 minutes	Students test their science communication plan from Module 4.2 with the chatbot.
Putting pieces together	5 minutes	Students gather the feedback from the chatbot and discuss what is interesting, unusual, or useful to revise their science communication plan.

MATERIALS

- [Lesson 4.3 slides](#)
- [Science journal](#)
- [Teacher Rubric](#) for science communication products (referenced throughout module 4)
 - [Student Rubric Checklist](#) (provided in science journal, p.60)
- [AISciComm chatbot website](#)

BEFORE YOU START TEACHING

1. Review [AISciComm chatbot website](#) to model and/ or support students.
2. Review the Science Communication Product [Student Checklist](#) (also in Science Journal p. 60) and [Teacher Rubric](#) to know the expectations of students.
3. Throughout the program, students will be working in teams of three (preferably).
4. Make sure that you have a way to project a video with sound.
5. The science journal is intended to be a digital notebook. However, if you decide to print copies for students, they may need a writing instrument (pen, pencil, etc.).

WHERE WE ARE GOING AND NOT GOING

Where We Are Going: Students will test their science communication plans with an audience simulation (chatbot) to see how well their message lands. They practice adapting their messages to a specific audience's needs, values, and background knowledge. They learn that **science communication is iterative**. Additionally, students will now apply their prompting skills with generative AI from prior modules to develop their own GenAI chatbots. Through this iterative process, they will gain a stronger understanding of how the technology works.

Where We Are NOT Going: We are not finalizing our products yet. This is still a testing and learning phase. Students should **not focus on building the perfect chatbots**. The chatbots are just tools to simulate audience reactions.

LEARNING SEQUENCE

Lesson plan notes are also provided on the slide notes.

NAVIGATE: Introduction to the Lesson (10 minutes)

1. Open the slideshow to **slide 1** and introduce the lesson. Explain to students that now that we have planned our messages and identified our target audience, **they'll build a GenAI chatbot to represent their chosen audience and help them test their message**.
2. Advance to **slide 2** and ask students to brainstorm: *What aspects of our target audience do we want to simulate, so that we can test our messages?*
 - a. Students can come up with responses like demographics (e.g., age, gender, occupation), interests, values they associate with the ocean, etc.
3. Advance to **slide 3** and walk students through a prompt example that contains different characteristics of a simulated profile:
 - a. **Role** (e.g., civil engineer)
 - b. **Task** (e.g., conversation about climate change)
 - c. **Background information** (e.g., living experiences in OC)
 - d. **Constraints** (e.g., when to cite sources),
 - e. **Example responses** (e.g., question-answer pairs so the chatbots can sound like our intended character)

INVESTIGATE: Build Prompts for our Chatbots to Test our Message (20 minutes)

1. Advance to **slide 4**. Explain to students that in their project teams, students will work on their science communication plan and develop the prompt for the chatbots to test their plans. Invite students to pull up the target audience profiles they created in Module 4.2.

2. Students brainstorm their “target audience” chatbot profile based on the following guiding questions:
 - a. **Role:** *What role will your chatbot play?* It could be another occupation (not already included in Module 3) or a local community figure. It would be best to include roles that you have encountered in your research in prior modules (e.g., interviewing community members, families, teachers, and friends).
 - b. **Task:** *What is the task that your chatbot should perform?* It could be: “critique my science communication plan” or “act as a conversational partner and ask me follow-up questions about my message”.
 - c. **Background information:** *Who is your chatbot?* You can build this part based on your answers in lesson 4.2 (Target audience).
 1. Age: [Specify age range]
 2. Race/Ethnicity: [Describe the racial/ethnic background]
 3. Living Experiences: [Urban, rural, coastal, etc.]
 - d. **Constraints:** *Are there any constraints you want your chatbot to do, so that its responses will be what you want?* For example: only answer questions related to science communication; only answer in 2-3 sentences; answer in fewer than 100 words, etc.
 - e. **Example interactions (OPTIONAL):** These chatbots work better if we provide them with example interactions (e.g., pairs of question-answer) of what the real person will sound like. You could add 2-3 example question-answer pairs here to provide examples for the chatbot.

INVESTIGATE: Test your Science Communication Message (15 minutes)

1. In project teams, students will input the prompt they have just created to create their target audience to the **AISciComm chatbot site > default chatbot**. Advance to **slide 5** to show a possible prompt ([Science Journals](#), p. 69). The prompt format could be:

You are role-playing as {character}. Here are details about you: you will act as this role: {role}. In this task, I want you to {provide task description}. You will role-play as an individual with this background information: {describe background, like age, race/ethnicity, living experiences, occupation}.

Your response should have the following constraints: {constraints}.

I will share details about a science communication product with you, and you will critique my plan from the perspective of this character.

Here is my product type: {product type, like a social media post, board game, or infographic}.

Here is my science communication plan: {plan}.

What do you think about this plan? Ask me questions about what is unclear or can be improved about my plan.

2. Advance to **slide 6** and show students an example conversation using this template. Ask student teams to test their communication plan and **take screenshots of the conversation with the chatbots roleplaying as the target audience. Add the screenshots to Google Drive. Instructions are on Slide 7.**
 - Remind students to **critically evaluate the chatbots' responses** and **ask follow-up questions and prompts** to revise the persona as needed.
 - Instructions to take a screenshot for students **slide 7**:
 - Devices will differ in how to take a snapshot.
 - If on a Windows computer
 - Press **Window + Shift + S** at the same time
 - If on a Mac computer
 - Press **Command (⌘) + Shift + 4** at the same time
 - If on a Chromebook
 - Press **Ctrl + Shift + Show Windows** at the same time
3. Advance to **Slide 8**. Remind students that they may use the **student science communication product checklist** as another helpful resource to ask the chatbot additional questions related to specific parts of their product: language and clarity, science accuracy, real-world and personal connections, structure, style, visuals, and audience engagement.
 - a. Provide students with an example, using the Lotería Game designed for middle school students. Ask students questions about this example:
 - i. **Structure:** *Does it have a logical structure to play the game?*
 - ii. **Style and visuals:** *Are there imagery, metaphors, or storytelling elements that could enhance the scientific content?*
4. Advance to **Slide 9** (similar to slide 5) so students can see the prompt template.

PUTTING PIECES TOGETHER: Add to Science Communication Plan (5 minutes)

1. Advance to **Slide 10**. Invite student teams to discuss the chatbot's responses to improve their science communication plan. See [Science Journal p. 71](#) for reference.
2. Students complete their individual science journal and respond to the questions: *What feedback about your science communication plan did you learn? Was there anything interesting or surprising?*